
Financial Spreading

*Turning raw borrower financials into spreads a credit
approver can trust*

PRODUCT REQUIREMENTS · COMMERCIAL CREDIT

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Executive Summary

What the Financial Spreading Engine automates — and where it stops

Field	Value
Product area	Commercial Credit — Automated Financial Spreading
Document owner	Eshwar Belani
Audience	Engineering, Product, Design, Credit/Underwriting SMEs
Status	Draft v0.4 — for team review
Loan domains	Commercial & Industrial (C&I) — MVP; Commercial Real Estate (CRE) — post-MVP, multifamily first
Requirement convention	FR-<AREA>-<n>, tagged P0 (must), P1 (should), P2 (could)
Reference packages	Regional Bank (RB) — real customer, priority (identity masked); YETI Holdings; MAA; CFI cases (RockCrusher, Rocky Mountain/Fisher, Woodchucks)
Methodology references	RMA Guide to Spreading Financial Statements (2022); CREFC Investor Reporting Package (OSAR / NOIWS); Fannie Mae / Freddie Mac multifamily standards; ParseBench (arXiv 2604.08538)

Financial spreading is the process by which a lender takes a borrower's raw financial information — tax returns, audited/reviewed/compiled statements, company-prepared and interim statements, 10-Ks, rent rolls, debt schedules, AR agings, borrowing-base certificates, guarantor personal financial statements — and re-expresses it in a standardized template so the institution can compare borrowers like-for-like, trend across periods, and compute the debt-capacity and risk metrics that drive a credit decision. Spreading is not “OCR into Excel”; it is policy-controlled, auditable credit analysis.

This product automates that end-to-end: ingest a loan package (a folder of documents, plus the customer's spreading template and a special-instructions file), classify and parse every document, map extracted data to a standardized chart of accounts, build period-aligned spreads (including interim and last-twelve-months views), apply lender-specific normalization and policy, and compute both standard and underwriter-defined custom metrics — all with full traceability back to the source so a credit approver can trust and audit every number. The MVP targets C&I; CRE follows, multifamily first.

Three realities shape the design. First, parsing is the floor: commercial-lending documents are among the hardest artifacts to parse (multi-year tables with merged cells and hierarchical headers, footnotes that change the numbers above them, image-only PDFs), and a misread figure does not look wrong — it flows silently into EBITDA, leverage, DSCR, and the LTM math. Second, spreading is not transcription: bank spreads deliberately and conservatively deviate from GAAP, and the rules are policy-specific. Third, every customer is different: underwriters build proprietary custom metrics and hand us their templates and formulas, expecting us to source the inputs from the raw package — including inputs that appear on no statement — and compute their way. Our bar is “reliable enough to act on,” not “good enough to read,” and a confident wrong answer is worse than a flagged uncertain one. The details are the product.

THE CENTRAL COMMITMENT

Spreading is policy-controlled, auditable credit analysis — not OCR into Excel. The engine automates the full lifecycle (ingest → classify → parse → map → spread → period/LTM → adjust → compute standard and custom metrics → validate → review → export) and carries a source citation and a High/Med/Low confidence tier on every delivered value. Accuracy is never traded for automation rate: no value is written silently, and a confident wrong answer is worse than a flagged uncertain one.

What this document establishes

- Ingest a C&I loan package (folder + template + instructions) and produce a standardized, multi-period spread with minimal manual entry.
- Map raw data to a configurable standard chart of accounts encoding the institution's spreading policy (including conservative GAAP deviations), driven primarily by the customer's template.
- Correctly construct interim and LTM periods from partial-year data, assurance-aware (the stub-period roll-forward).
- Apply configurable adjustments/normalization with documented rationale and aggressive cross-period consistency checks.
- Compute the standard ratio catalog plus lender-defined custom metrics (including covenant tests).
- Produce a clean Excel export with confidence, provenance, source citations, and dedicated commentary worksheets.
- Provide full traceability (every value → transformation → source document/page/region) and a review/approval workflow.
- Let institutions encode their spreading SOP/policy as versioned configuration, overridable per package by the special-instructions file.

What this document does NOT cover

This document specifies the spreading engine, not the credit decision. Adjacent concerns are out of scope for the initial release:

- ▶ Rendering a credit decision or risk rating — we feed these; the human/policy decides.
- ▶ Originating, documenting, or servicing loans; loan accounting.
- ▶ Forecasting/projection modeling as a primary capability (we may populate a projection template; driver-based forecasting is P2).
- ▶ Tax-return preparation or any attest function; pass-through/entity-level tax handling is deferred to V1.
- ▶ Parsing covenant definitions out of loan agreements (MVP supports configured covenant metrics; agreement parsing is later).
- ▶ Asserting covenant compliance unless the covenant definition is configured and user-validated.

Change log and notes

Change log. v0.4 locks the open-question decisions: C&I is the MVP; CRE is post-MVP (multifamily first); a per-package input model of documents folder + spreading template + special-instructions file; MVP default templates = RMA C&I chart of accounts + tax-return structure (CRE defaults — CREFC OSAR/NOIWS + agency multifamily — ship with the CRE phase); a clean Excel export with dedicated commentary worksheets and High/Med/Low confidence surfaced via cell notes, a subtle traffic-light tier, provenance styling, an Exceptions sheet, and source links; covenants treated as configured custom metrics (loan-agreement parsing deferred); pass-through/tax handling deferred to V1; aggressive inconsistency flagging; and source citations on every delivered value. All borrower PII is masked. v0.3 verified RB facts and added tax/relief/provenance items; v0.2 added the assurance engine, validation/severity, and measurement; v0.1 established the core pipeline, custom-metrics engine, and worked LTM example.

Note on borrower anonymity. The priority reference is a real customer's borrower; its identity and entity names are masked throughout (“the RB borrower,” “an affiliate VIE”). De-identified figures are retained only because they make the requirements concrete and are not identifying once names, industry, and location are removed.

Reading-order note for the team. Before building against the RB package, read the right-hand “Calculations” columns of RB - Output Cleaned.xlsx. They encode the LTM roll-forward and the custom EBITDA/EBITDAR/fixed-charge logic line-by-line and are the clearest statement of what real customers expect. Appendix D gives the corrected formula — the human-prepared note in that file contains a sign-labeling inconsistency the engine must not reproduce.

2. Background and problem statement

2.1 What spreading is (and what makes it hard)

RMA: WHAT SPREADING IS

“Spreading is the process by which a bank transfers information from a borrower's financial statements into the bank's financial analysis spreadsheet program.” A correct spread generates common-size statements, financial ratios, a statement of cash flow, and a reconciliation of net worth. Member banks submit spreads into RMA's Annual Statement Studies®, which is why a standard, uniform method matters — spreads that consistently beat or trail industry medians can indicate a methodology problem, not a strong or weak borrower.

The difficulty is that good spreading deliberately departs from how the financials are presented:

- Bank spreads deviate from GAAP, on purpose, conservatively. Prepaid expenses → non-current; restricted/sinking-fund cash → non-current; cash surrender value of life insurance → non-current; deposits and long-term deferred charges → non-current; receivables due from related parties → non-current (discretionary repayment); long-term debt owed to related parties → current unless explicitly subordinated to the bank.
- Statement quality doesn't change the rules but it changes confidence: audited > reviewed > compiled > company-prepared > tax-basis. The same conservative rules apply, but a compilation or internal interim warrants lower trust and may even omit a statement (see §2.4).
- Line items hide in multiple places. D&A across COGS and opex; interest buried in COGS (homebuilders); officers' compensation inside SG&A. Footnotes and schedules routinely change the spread — they carry related-party balances, debt terms, contingent liabilities, litigation, and even obligations disclosed but not yet recorded.
- Exceptions are inevitable and must be explained. RMA: spreading “should not be a mechanical exercise on autopilot... If an exception is made, the rationale should be well supported and explained so the Credit Approver is aware of the change.”

2.2 Why custom metrics matter

In both C&I and CRE, underwriters build custom metrics — their own EBITDA definitions, coverage tests, leverage variants, debt-yield and borrowing-base formulas — captured in shared templates, because a proprietary, consistent lens is a competitive underwriting edge. The product must let lenders define, configure, and govern their own metrics, and reliably gather the underlying inputs from the raw package — including inputs that are derived, assumed, or sourced from footnotes and schedules rather than appearing as a clean line item.

The real RB customer spread is the canonical example: a house EBITDA labeled “CIT EBITDA” (Gross Profit – SG&A – Rent), an Adjusted EBITDA that adds back specific items (e.g., bad-debt expense), an Adjusted EBITDAR that further adds back rent, leverage tests of Total Debt / CIT EBITDA, Total Debt / Adjusted EBITDA, and (Debt + Leases) / EBITDAR, and a multi-step Adjusted Fixed Charge Coverage that splits capex into maintenance vs. growth, layers in financed capex and lease expense, and divides by cash interest plus principal repayments. Several inputs (maintenance-vs-growth capex split, financed capex) are assumption-driven and not directly disclosed — the spread literally annotates “not clear how these numbers have been obtained.” The product must support exactly this.

2.3 The parsing foundation

SMALL PARSING ERRORS BECOME DECISION ERRORS

A large share of source and even reference documents are image-based PDFs (the MAA spread, several YETI spreads, and the RB credit-app exhibit are images, not selectable text). Extraction quality is the floor under everything else. We hold parsing to a benchmark standard (ParseBench-style: tables, charts, content faithfulness, semantic formatting, visual grounding), because mistranscribing a 2.0% rate as 0.2%, or shifting a merged-cell header, corrupts the model while failing silently.

2.4 Three concrete failure classes the product must catch

- Parsing-induced silent error. A single misread cell in a merged-header, multi-year table propagates into EBITDA → leverage → covenant headroom → the credit grade, with nothing on the page flagging it. This is why we confidence-gate, run internal tie-outs, and keep visual grounding.
- Judgment / cross-period inconsistency. In the RB working spread, 2022 bad-debt expense of \$3,662,933 (5.96% of revenue) is added back to EBITDA as a non-recurring adjustment, while the “Adjustments” line is zero in every other year — even though 2021 included a \$380,492 bad-debt charge (a customer's bankruptcy, disclosed in the notes) and 2023 bad debt was only ~\$19K. Whether or not the one-year add-back is correct, the engine must surface this cross-period inconsistency, not silently reproduce it.
- Footnote-driven, not-on-the-face items. The RB audit discloses a ~\$573K tax obligation that is not recorded on the balance sheet, plus a \$2,331,124 relief refund and related-party leases — none visible from the face of the statements, all changing the credit picture. The engine must read footnotes and surface such items.

2.5 Current state / pain

Spreading today is overwhelmingly manual: analysts re-key multi-year statements into Excel or a spreading platform, hand-apply reclassification and add-back policy, read footnotes for related-party

balances and buried interest, hand-build interim/LTM columns, compute custom metrics, and a second reviewer ties it out. It is slow, inconsistent across analysts, hard to audit, and the errors are silent. The opportunity is consistency, speed, and auditability without taking away the underwriter's judgment or their custom lens.

3. Goals, non-goals, and success metrics

3.1 Goals

- Ingest a C&I loan package (folder + template + instructions) and produce a standardized, multi-period spread with minimal manual entry.
- Map raw data to a configurable standard chart of accounts encoding the institution's spreading policy (including conservative GAAP deviations), driven primarily by the customer's template.
- Correctly construct interim and LTM periods from partial-year data, assurance-aware (the stub-period roll-forward).
- Apply configurable adjustments/normalization with documented rationale and aggressive cross-period consistency checks.
- Compute the standard ratio catalog plus lender-defined custom metrics (including covenant tests).
- Produce a clean Excel export with confidence, provenance, source citations, and dedicated commentary worksheets.
- Provide full traceability (every value → transformation → source document/page/region) and a review/approval workflow.
- Let institutions encode their spreading SOP/policy as versioned configuration, overridable per package by the special-instructions file.

3.2 Non-goals (initial release)

- ▶ Rendering a credit decision or risk rating (we feed these; the human/policy decides).
- ▶ Originating, documenting, or servicing loans; loan accounting.
- ▶ Forecasting/projection modeling as a primary capability (we may populate a projection template; driver-based forecasting is P2).
- ▶ Tax-return preparation or any attest function; pass-through/entity-level tax handling is deferred to V1.
- ▶ Parsing covenant definitions out of loan agreements (MVP supports configured covenant metrics; agreement parsing is later).
- ▶ Asserting covenant compliance unless the covenant definition is configured and user-validated.

3.3 Success metrics

Metric	Target (directional)
Straight-through spreads requiring no manual correction	≥ 70% of standard C&I packages by GA
Field-level numeric extraction accuracy on financial tables	Benchmark-gated; 100% of sub-threshold cells flagged (no silent low-confidence values)
Tie-out pass rate (balance control, cash-flow tie, equity roll-forward)	100% pass-or-flagged
Custom-metric fidelity vs. customer ground-truth template	Within rounding on every reference package
Traceability	100% of delivered values carry a source citation and a confidence tier
LTM correctness on reference cases	100% (Appendix D as a regression test)
Time to first complete spread	Hours of analyst work → minutes

4. Users and personas

- Credit Analyst (primary “spreader”). Runs the spread, resolves extraction/classification exceptions, applies add-backs, builds the interim/LTM view.
- Underwriter / Relationship-side credit. Owns the borrower view and the custom metrics; needs their proprietary lens reproduced exactly and “why is this number this?” answerable instantly.
- Credit Officer / Approver. Consumes the spread; must see rationale for every deviation/adjustment and trace any number to source; signs off.
- Credit Policy / Risk Administrator. Configures the standard chart of accounts, deviation rules, add-back policy, metric library, thresholds, and workflow.
- Spreading Operations Manager / Shared Service. Owns the queue, SLAs, and reviewer assignment; needs throughput, consistency, exception routing.
- Portfolio Manager. Ongoing monitoring, periodic re-spreads, covenant headroom, trend continuity.
- Model Risk / Validation & Internal Audit; External Examiner. Need documented rules, versioned policies, complete audit trail, reproducibility, and source traceability.
- Implementation / Customer Success. Configures new lenders, imports templates, validates historical spreads, migrates spreadsheet logic.

5. Scope and per-package inputs

In scope: automated spreading for C&I (MVP) and CRE (post-MVP, multifamily first) borrowers and their guarantors; privately held companies, public filers (10-K/10-Q), real-estate entities/SPEs, holding-company structures with subsidiaries/affiliates/VIEs; audited/reviewed/compiled/company-prepared/tax-basis statements; historical multi-year, interim/stub, and LTM periods; standard + custom metrics; SOP/policy configuration; traceability, workflow, export/integration.

Out of scope (initial): consumer/retail lending, full projection engines, covenant negotiation, automated decisioning, document generation, non-USD consolidation, entity-level tax handling (V1).

5.1 Per-package inputs (MVP)

Each spreading job is defined by three inputs supplied together:

- A documents folder — all required source documents for the borrower (statements, tax returns, interims, debt schedule, agings, guarantor PFS, etc.). MVP assumes the folder is complete; the completeness/freshness checks are V1.
- A spreading template — the customer's target format: the chart of accounts / line structure, the reclassification conventions, and the custom-metric and covenant formulas. This is the primary driver of how the spread is built; the engine parses it (labels and formulas) and reproduces it with live formulas.
- A special-instructions file — per-package guidance from the end user, including the source-precedence ranking (which document/period wins when sources disagree) and any other directives (preferred periods, add-back guidance, which metrics to run, masking, thresholds). Instructions override Policy-Profile defaults for that package.

5.2 Reference-package mapping to scope

Package	Domain	What it tests
Regional Bank (RB) — priority (borrower masked)	C&I services (privately held operating company; S-corp)	Real customer; stub-period LTM roll-forward; custom CIT-EBITDA/EBITDAR + fixed-charge tab; GAAP-deviation reclassifications; consolidated group with an affiliate VIE; mixed assurance; footnote-driven items (relief refund, unrecorded tax obligation)
YETI	C&I / ABL	Public 10-Ks; borrowing-base certificate (eligibility/advance rates/reserves); multiple output

Package	Domain	What it tests
		formats incl. machine-readable Moody's-style reported/adjustment/adjusted
MAA	CRE / multifamily REIT	Entity-level REIT spread (EBITDAre, net debt); image- based spread requiring OCR/vision; appendix tying to 10- K pages
Rocky Mountain / Fisher	CRE / property-level	Commercial-mortgage model: rent roll, normalized NOI, DSCR/LTV/debt yield, cap-rate sensitivity, tenant/lease analysis, risk-rating scorecard
CFI cases (RockCrusher, Woodchucks)	Mixed / training	Raw inputs + blank bank-style templates — structure references, not ground-truth targets

6. Product overview — the spreading lifecycle

The engine is a pipeline; each stage is independently testable, re-runnable, and auditable, and a correction at any stage re-propagates downstream:

- Ingest the folder + template + instructions.
- Classify documents, entities, and periods (with assurance level).
- Extract/parse with source coordinates and confidence.
- Normalize & map to the template's chart of accounts.
- Apply spreading rules (policy-driven GAAP deviations; instructions override).
- Resolve periods (annual, interim, LTM; assurance-aware).
- Adjust/normalize (add-backs, candidates vs. approved).
- Compute standard + custom + covenant metrics.
- Validate & tie out (aggressive flagging; confidence/severity).
- Human review/approve with exceptions and rationale.
- Export the clean workbook (+ commentary tabs) and integrate.

DESIGN TENETS

No silent low-confidence values; source coordinates retained from extraction to export; every value carries a provenance status and a confidence tier (High/Med/Low); the Policy Profile is the versioned configuration spine, overridable per package by the instructions file; a human can override anything, with author, timestamp, rationale, and the superseded value captured.

7. Functional requirements

7.1 Ingestion: folder, template, and instructions (FR-ING)

- FR-ING-1 (P0). Accept a documents folder as the package (PDF, scanned/image PDF, XLSX/XLSM/CSV, DOCX) including multi-document bundles (a single PDF holding several statements/years). Preserve originals immutably. MVP assumes the folder is complete.
- FR-ING-2 (P0). Ingest the spreading template: parse its line labels, chart-of-accounts structure, reclassification conventions, and formulas (standard ratios, custom metrics, covenant tests); map its lines to the SCOA; treat it as the primary specification of the target spread (FR-CUS-3). Detect hardcoded plugs masquerading as formulas and flag them (FR-EXT-8).
- FR-ING-3 (P0). Ingest the special-instructions file: the source-precedence ranking and any other per-package directives; apply them as overrides on the Policy Profile for this package (FR-SRC-2, FR-SOP-1).
- FR-ING-4 (P0). Auto-classify each document into a taxonomy (Appendix E): audited/reviewed/compiled statement; company-prepared/internal interim; 10-K/10-Q; business tax return (1120/1120-S/1065/Sch C/K-1); credit application; debt schedule; AR/AP aging; inventory report; borrowing-base certificate; covenant compliance certificate; rent roll; operating statement (T-12/YTD); appraisal; guarantor PFS; personal tax/W-2; org/ownership docs; completed-spread exhibit. Surface confidence; low-confidence items queue for confirmation.
- FR-ING-5 (P0). Split bundles into constituent documents and periods, including multiple fiscal years in one file and multiple statements (BS/IS/CF/notes) per document. (The RB audits each carry two comparative years plus notes; the engine must separate them.)
- FR-ING-6 (P0). Identify per document: borrower/entity, statement basis (GAAP/tax/cash), assurance level, reporting currency/scale (“\$ in thousands”), fiscal year-end, and period type (full-year vs. interim/stub) with cutoff date.
- FR-ING-7 (P1). (V1) Package-completeness report against a checklist and a freshness check for stale interims; placeholder/empty-file detection.

7.2 Data extraction and parsing (FR-EXT)

- FR-EXT-1 (P0). Extract financial tables (BS, IS, CF), labels, values, period headers, and units from both digital-text and image-based PDFs (OCR/vision) with equal fidelity. Image extraction is mandatory.
- FR-EXT-2 (P0). Preserve table structure: merged cells, spanning/hierarchical headers, indentation hierarchy (subtotal vs. component), repeated/continued headers, multi-column layouts.
- FR-EXT-3 (P0). Normalize numeric representation: parenthetical / trailing-minus / red-text negatives; thousands/millions scaling; and distinguish “—”/blank meaning zero from genuinely missing.
- FR-EXT-4 (P0). Extract footnotes, notes, and schedules and bind them to the line items they qualify (related-party, concentration, debt, line-of-credit). Capture disclosed-but-unrecorded items — contingent liabilities, litigation, subsequent events, and obligations not on the face of the statements (e.g., the RB ~\$573K unrecorded tax obligation and the \$2.33MM relief refund). RMA: “the footnotes must be read.”
- FR-EXT-5 (P0). Retain source coordinates for every value (file, page, table, bounding box, raw text; for Excel: sheet + cell + formula). These persist to the export (visual grounding; FR-AUD, FR-OUT).
- FR-EXT-6 (P0). Emit multi-level confidence (document-class, table-detect, header/period, value, sign, unit, mapping, formula-input, metric-output), summarized to a High/Med/Low tier per delivered value. Any value below threshold is flagged and never written silently; confidence is explainable (a Low-confidence EBITDA traces to its weakest input).
- FR-EXT-7 (P0). Internal identity checks at extraction: a stated subtotal vs. the sum of its parts; flag mismatches as probable OCR/extraction errors before they propagate.
- FR-EXT-8 (P0). For Excel sources (incl. the customer template), extract values and formulas, sheet structure, named ranges, comments, and hidden tabs, and detect hardcoded values/hidden assumptions (a number typed over a formula, an off-sheet constant) — surfaced, not trusted blindly.
- FR-EXT-9 (P1). Extract structured form documents by field (borrowing-base certificate, covenant certificate, PFS).
- FR-EXT-10 (P1). For 10-K/10-Q, locate consolidated statements, notes, MD&A, non-GAAP reconciliations, and segment data.
- FR-EXT-11 (P1). Extract chart/figure values where financials are presented graphically (lower-confidence; flag).

7.3 Standardization and mapping (FR-MAP)

- FR-MAP-1 (P0). Map to the customer template's chart of accounts; where no template line exists, fall back to a configurable standard chart of accounts (SCOA) — MVP default is RMA-aligned, plus a tax-return-structured layout for tax-based borrowers.

- FR-MAP-2 (P0). Map each extracted line to a target line, normalizing label/placement variation across accountants and years (e.g., “Revenue from services” / “Net sales”; “Operating line of credit” / “Revolver”). Support one-to-many and many-to-one mappings, rollups, and drill-down.
- FR-MAP-3 (P0). Encode policy-driven reclassifications that intentionally deviate from GAAP, defaulted to RMA guidance (Appendix A): prepaids, restricted/sinking-fund cash, CSVLI, deposits, deferred charges → non-current; related-party receivables → non-current; third-party notes receivable split by maturity; related-party debt → current unless subordinated; CPLTD broken out as current.
- FR-MAP-4 (P0). Support break-outs of items hidden inside other lines: officers'/owner comp out of SG&A; D&A across COGS and opex; interest embedded in COGS; split “interest expense and amortization of debt costs” into components where formulas need it.
- FR-MAP-5 (P0). Preserve as-reported vs. as-spread as distinct layers — a reclassification never destroys the as-reported value; it adds a layer with its own lineage.
- FR-MAP-6 (P0). Every mapping/reclassification is inspectable, overridable, and logged with rationale. Learn and persist borrower/industry-specific mappings from accepted corrections without overriding policy-defined mappings.
- FR-MAP-7 (P1). Support layered presentations of the same underlying spread: Reported → Normalized → Bank-adjusted → Adjustment candidates → Underwriter-approved adjusted → Covenant → Pro forma → Projection. Reported is immutable; candidates cannot flow into official/covenant metrics unless approved.
- FR-MAP-8 (P1). Map to SCOA categories compatible with RMA Annual Statement Studies for peer benchmarking.

7.4 Statement quality, assurance, and source precedence (FR-SRC)

- FR-SRC-1 (P0). Detect and label assurance level per statement/period (audited / reviewed / compiled / tax / company-prepared / internal interim / management schedule / derived). Surface it prominently.
- FR-SRC-2 (P0). The end user sets source precedence. The ranking comes from the special-instructions file supplied with the package (FR-ING-3); the engine applies it when two sources cover the same period and records which source won. A configurable default hierarchy (audited > reviewed > compiled > tax > company-prepared > internal interim > management schedule > derived) is used only as a fallback when instructions are silent. A lender may also invert it (e.g., prefer tax returns for small-business income).
- FR-SRC-3 (P0). Build periods from the chosen source: anchor on the highest-precedence full year and use the most recent interim for the current/LTM view; carry each component's assurance into the resulting period.

- FR-SRC-4 (P1). Flag a stale spread (relies on older periods than newer available data) and flag when a lower-assurance interim feeds the most decision-relevant column.
- FR-SRC-5 (P0). Worked profile (RB), verified against the borrower's audited statements (identity masked): 2018/2019 reviewed; 2020/2021 audited; 9-mo 2022 compilation (omits the statement of cash flows and substantially all disclosures); 9-mo 2023 internal/company-prepared; 2022/2023 audited. The LTM Sept-2023 column blends an audited prior-year anchor, a compiled prior stub, and an internal current stub — the engine must reflect that mixed assurance and the cash-flow gap (FR-PER-8).

7.5 Spreading the statements (FR-SPR)

- FR-SPR-1 (P0). Produce the standardized balance sheet (current/non-current assets, current/non-current liabilities, intangibles, net worth) reflecting all FR-MAP reclassifications, with common-size columns.
- FR-SPR-2 (P0). Produce the standardized income statement (revenue, COGS, gross profit, opex detail, operating income, non-operating/one-time items, taxes, net income) with common-size and YoY growth.
- FR-SPR-3 (P0). Produce a statement of cash flow / UCA (“bankers”) cash flow and a reconciliation of net worth / changes in equity across periods.
- FR-SPR-4 (P0). Recompute everything automatically when any upstream value or rule changes; never present stale metrics.

7.6 Period management and interim / LTM construction (FR-PER)

High-risk, high-value: this feeds EBITDA, leverage, and coverage on the most recent and most decision-relevant period.

HOW LTM IS CONSTRUCTED

Flow items (income statement, cash flow): $LTM = \text{Current interim (YTD)} + (\text{Prior full year} - \text{Prior-year interim of equal length})$. Example: $LTM \text{ Sep-2023} = \text{CY 9/23} + (\text{FY2022} - \text{PY 9/22})$. The bracket is the prior year's non-overlapping tail (Oct–Dec 2022). Stock/balance-sheet items: LTM uses the most recent period-end balance, never a roll-forward. Verified: $\text{Revenue } 57,581 + (61,443 - 44,326) = 74,698$.

- FR-PER-1 (P0). Represent each period with explicit attributes: fiscal year-end, start/end dates, length in months, full-year vs. interim/stub, flow vs. stock applicability, and (for LTM) its construction method and the assurance of each component.
- FR-PER-2 (P0). Align periods across statements with differing fiscal year-ends onto one timeline and label them consistently (FY2019–FY2022, PY 9/22, CY 9/23, LTM Sept 23).

- FR-PER-3 (P0). Construct LTM for flow items (income statement, cash flow) line-by-line: $LTM = \text{Current interim (YTD)} + (\text{Prior full year} - \text{Prior-year interim of equal length})$, i.e., $LTM \text{ Sep-2023} = \text{CY 9/23} + (\text{FY2022} - \text{PY 9/22})$. Compute line-by-line because not every line exists in every period (Appendix D).
- FR-PER-4 (P0). For stock/balance-sheet items, LTM uses the most recent period-end balance (the Sept-2023 balance), never a roll-forward. (The Sept-2023 interim balance differs from the FY2023 year-end audit — interim $\neq$ year-end.)
- FR-PER-5 (P0). Guardrails: require matching fiscal cutoff months and equal interim lengths before computing a flow LTM; if the prior-year interim is missing, do not fabricate — flag as uncomputable. Warn on mismatched calendars and irregular/short years.
- FR-PER-6 (P0). Do not transcribe human notes as logic. The reference workbook's note reads “CY – (FY – PY),” which is the wrong sign though its computed cell is correct; the engine encodes the verified formula and shows the derivation. Permanent regression test (§9.3).
- FR-PER-7 (P0). Label every period column with its method, basis, and assurance (e.g., “LTM Sep-2023 = 9-mo 2023 + (FY2022 – 9-mo 2022); current stub company-prepared”).
- FR-PER-8 (P0). Cash-flow-gap handling. If an interim omits the statement of cash flows (e.g., the RB 9-mo 2022 compilation), compute income-statement LTM but flag cash-flow- and fixed-charge-derived LTM metrics as requiring an explicit, approved derivation or additional schedules — do not silently tie them out.
- FR-PER-9 (P1). Support annualization/TTM as configurable, explicitly-labeled alternatives (not for covenant testing unless policy allows); support 13-week, 52/53-week, fiscal-month, and quarter-end calendars.
- FR-PER-10 (P1). Apply LTM consistently to derived custom metrics: an LTM metric is computed from LTM inputs, not by bridging the ratio itself.

7.7 Adjustments, add-backs, and normalization (FR-ADJ)

- FR-ADJ-1 (P0). Compute EBITDA and configurable variants (EBITDAR, house definitions) by adding back interest, taxes, D&A to operating income, each add-back itemized and sourced, shown as a transparent build (reported → adjustments → adjusted).
- FR-ADJ-2 (P0). Maintain a configurable add-back library: non-cash (D&A, impairment, stock comp), non-recurring (litigation, gains/losses on asset sales, product recall), discretionary owner comp / personal expenses, lease/rent (→ EBITDAR), storm/one-time repairs, financed capex. Each carries amount, period(s), source/derivation, recurring vs. non-recurring tag, approval status, and rationale; nothing is added back silently.

- FR-ADJ-3 (P0). Distinguish adjustment candidates from lender-approved add-backs. Disclosed unusual items (the YETI “product recall” and “impairment”) are shown for review and are explicitly not approved add-backs by default until policy/approver authorizes them.
- FR-ADJ-4 (P0). Cross-period consistency check (aggressive). If an add-back/adjustment is applied in one period, flag inconsistent treatment of the same line in all other periods unless a documented reason exists — directly targeting the RB bad-debt case (§2.4). Expressible in the custom-metric DSL (FR-CUS-2).
- FR-ADJ-5 (P0). Support a house EBITDA definition that differs from textbook EBITDA — e.g., RB's “CIT EBITDA” = Gross Profit – SG&A – Rent — and rent/lease add-backs to derive EBITDAR.
- FR-ADJ-6 (P0). Apply caps/conditions from policy/instructions (e.g., owner-comp add-back capped; bad-debt added back only if flagged non-recurring), evaluated automatically with exceptions surfaced.
- FR-ADJ-7 (P0). Government-relief and other one-time income. Recognize and tag relief/one-time items — relief refunds (the RB \$2,331,124), PPP forgiveness, grants, insurance recoveries, gains on asset sales — as non-recurring, default them out of run-rate earnings, and show their effect explicitly.
- FR-ADJ-8 (P1) — V1. Pass-through / entity-level tax handling. Detect entity tax status (e.g., an S-corp with no federal/state income-tax provision; its “Income Taxes” line is immaterial franchise tax); model cash taxes via owner tax distributions for coverage and surface disclosed-but-unrecorded tax obligations (the RB ~\$573K). Deferred to V1 per scope.

7.8 Standard ratio and metric engine (FR-RAT)

FR-RAT-1 (P0). Compute and trend a standard ratio catalog, each with a transparent formula and drill-down:

Ratio	Measures	Formula (default)	Typical benchmark
DSCR	Cash flow vs. debt service	EBITDA (or CFADS) / annual debt service	> 1.20x–1.25x
Fixed-charge coverage	Coverage incl. leases/fixed costs	(EBITDAR – capex/taxes as configured) / (interest + principal + rent)	≥ 1.0x–1.25x
Current / quick ratio	Liquidity	Current assets / current liabilities; quick = (cash + trade AR) / current liabilities	> 1.0–1.2
Leverage (D/E)	Debt vs. equity	Total liabilities / tangible net worth	< 3.0x–4.0x

Ratio	Measures	Formula (default)	Typical benchmark
Funded-debt leverage	Debt vs. cash flow	Total funded debt / EBITDA	policy
Days receivable / inventory / payable	Working-capital cycle	(balance / sales or COGS) × days	industry-dependent
Margins	Profitability	Gross / EBITDA / net ÷ revenue	trend

- FR-RAT-2 (P0). Use tangible net worth (net of intangibles, and of disallowed related-party receivables where policy requires) and FR-MAP reclassifications, so liquidity and leverage reflect the conservative bank view.
- FR-RAT-3 (P1). Surface metric definitions inline; benchmark against RMA Statement Studies peers and flag material divergence.

7.9 Custom-metrics and covenant engine (FR-CUS) — centerpiece

Underwriters' proprietary metrics are a core requirement, and covenants are a special case of them (a custom metric + a threshold + a test period). The product must let institutions and underwriters define, compute, govern, and reuse their own metrics, sourcing inputs from the raw package — including inputs that are derived, assumed, or live in ancillary documents.

- FR-CUS-1 (P0). Provide a metric definition schema: name, description, category, formula, ordered inputs, period basis (FY/interim/LTM/TTM/point-in-time), entity applicability, unit, display format, polarity, threshold/covenant linkage, rounding rule, input-fallback behavior, test cases, version, and approval status. Versioned and attributable.
- FR-CUS-2 (P0). Provide a formula DSL with arithmetic; references to SCOA lines, standard metrics, and other custom metrics; multi-period references (prior period, LTM, average/CAGR of N years); conditionals (e.g., “add back bad debt only if flagged non-recurring” — making FR-ADJ-4 expressible); min/max/caps; aggregation across entities; unit- and sign-aware arithmetic. It must express the real RB metrics: CIT EBITDA, Adjusted EBITDA/EBITDAR, the leverage suite, and the multi-step Adjusted Fixed Charge Coverage (Appendix B).
- FR-CUS-3 (P0). Template ingestion is the MVP anchor. Parse the customer's template labels and formulas, map referenced lines to the SCOA, and reproduce the template with live formulas (input tabs feeding a formatted output tab) so the customer can audit and re-drive it. Surface (don't inherit) any hardcoded plugs (FR-EXT-8).
- FR-CUS-4 (P0). Covenant tests as configured custom metrics. Support covenant definitions entered/configured by the underwriter — a metric formula (often bespoke: a covenant “EBITDA” or “Funded Debt” that differs from the house version), a threshold, and a test period — and

compute pass/fail/headroom. Parsing covenant definitions out of loan agreements is deferred (V1+); MVP uses configured definitions (often carried in the template or instructions). See Appendix C.9 for why this lives in the spread.

- FR-CUS-5 (P0). Input sourcing beyond the statements. A metric input may point at an SCOA line, a standard metric, a footnote/schedule field, or a field from an ancillary document — AR aging (% over 90 days, concentration), inventory report, borrowing-base certificate (eligible AR, advance rate, reserves, availability), debt schedule (amortization, balloon, rate, cash interest, principal), or guarantor PFS. The engine locates and binds these from the package.
- FR-CUS-6 (P0). Assumption inputs are first-class. Where an input is not reported and must be assumed/allocated (the RB maintenance-vs-growth capex split; financed capex; cash interest vs. accrued; principal from the schedule/notes), mark it an assumption with an editable value and rationale, and never present it as a reported figure.
- FR-CUS-7 (P0). Each custom metric shows full derivation (formula, every input's value and source, periods, assumptions) — same standard as built-in ratios.
- FR-CUS-8 (P0). Ship a starter library (CIT/Adjusted EBITDA, EBITDAR, fixed-charge variants, lease-adjusted leverage, debt yield, borrowing-base availability, working-capital dependency, distribution coverage), each editable per institution.
- FR-CUS-9 (P1). Governance/scoping: metrics scoped global / institution / product / deal with inheritance/override; promotion personal → shared → policy-standard with approval; an “analysis-only” flag preventing flow into official covenant calculations unless approved; versioning recorded per spread.
- FR-CUS-10 (P1). Validation at definition and run time: undefined references, unit mismatches, circular references, division-by-zero, missing inputs — surfaced, never a silent zero/error.
- FR-CUS-11 (P2). Natural-language assist: describe a metric in prose → proposed structured definition + input bindings for human confirmation.

7.10 C&I-specific requirements (FR-CNI) — MVP focus

- FR-CNI-1 (P0). Working-capital analysis: current/quick ratio, net working capital, cash-conversion cycle (DSO/DIO/DPO), trends.
- FR-CNI-2 (P0). Debt-schedule & debt-service engine. Parse existing and proposed debt: lender, facility type (LOC, term, mortgage, equipment, SBA, lease), original/current balance, current vs. long-term portion, rate, payment, interest-only period, maturity, balloon, collateral, guarantees, subordination, prepayment, covenant linkage. Compute debt service; derive cash interest and principal for coverage.
- FR-CNI-3 (P0). Proposed-debt scenarios: recompute DSCR/FCCR/leverage under a new debt structure (sources & uses).

- FR-CNI-4 (P1). Borrowing-base & collateral quality: ingest borrowing-base certificates and AR/inventory agings; apply eligibility exclusions (>90-day, foreign, contra, affiliate, disputed, over-concentration; obsolete/slow-moving/consigned inventory), advance rates, reserves; compute eligible collateral and availability = borrowing base – outstanding – LC. Treat a public filer's indicative borrowing base (YETI) as illustrative until lender rules and field-exam results are loaded.
- FR-CNI-5 (P1). Funded-debt and lease-adjusted leverage consistent with add-back policy.
- FR-CNI-6 (P2). Projections / pro forma: populate a borrower projection template (e.g., the CFI RockCrusher driver-based 3-statement model) with the historical spread as the base; driver-based forecasting remains P2.

7.11 CRE-specific requirements (FR-CRE) — post-MVP (V1+, multifamily first)

The CRE phase begins after the C&I MVP and leads with multifamily (largest CRE debt segment). The Rocky Mountain/Fisher commercial-mortgage model is the property reference; MAA is the REIT reference. CRE default templates ship here: the CREFC OSAR (Operating Statement Analysis Report) and NOI Adjustment Worksheet (NOIWS) plus agency multifamily standards (Fannie Mae Form 4660 tiers; Freddie Mac Optigo).

- FR-CRE-1 (P1). Operating-statement spreading (3-year historical, YTD, T-12) into a standardized property-cash-flow format: gross potential rent, vacancy/credit loss, other income, EGI, opex by category, NOI, and NCF (NOI – replacement reserves – recurring capex – leasing costs).
- FR-CRE-2 (P1). Rent-roll spreading: suite/tenant, SF, base rent, escalations, recoveries/CAM, lease start/end, options, occupancy; compute physical and economic occupancy, WALT, in-place vs. market rent, rollover/tenant concentration; reconcile rent-roll annualized base rent to operating-statement rental revenue.
- FR-CRE-3 (P1). Normalized/underwritten NOI (Actual vs. Normalized columns), aligned to CREFC OSAR/NOIWS: vacancy floor (greater of actual or market), management-fee floor, replacement reserves (per-unit / per-SF), exclusion of one-time items, real-estate-tax normalization to the reassessed basis for acquisitions; in-place vs. stabilized NOI.
- FR-CRE-4 (P1). Core CRE metrics: DSCR (NOI or NCF / debt service), debt yield, LTV, LTC, cap rate, NOI per SF, break-even occupancy, expense ratio, and maximum supportable loan = $\text{NOI} \div (\text{min DSCR} \times \text{debt constant})$ at a test cap rate — with agency-multifamily thresholds (e.g., min DSCR 1.25x fixed; LTV ≤ 80%) configurable.
- FR-CRE-5 (P1). Appraisal integration: as-is / as-stabilized value, cap rate, comparables, valuation date/method; reconcile appraised vs. income-derived value; flag stale appraisal.
- FR-CRE-6 (P1). Tenant quality & lease-rollover analysis and a risk-rating scorecard (the Rocky Mountain model's structure).

- FR-CRE-7 (P1). Sensitivity / stress: vary cap rate, vacancy, and rate to show impact on value, LTV, DSCR.
- FR-CRE-8 (P1). REIT-style entity analysis (MAA): EBITDAre / adjusted EBITDAre, net debt, net debt / EBITDAre, debt / assets, debt / capitalization, FFO/AFFO, portfolio NOI.
- FR-CRE-9 (P2). Asset-type specializations beyond multifamily (office, retail, industrial, hospitality), sequenced by market size; construction/draw and lease-up.

7.12 Multi-entity, VIE, and related-party model (FR-CON)

- FR-CON-1 (P0). Model borrower structures (parent/subsidiary/affiliate/VIE/guarantor/property) with ownership percentages from org charts/notes; attach each statement to the correct entity and know what is consolidated vs. a separately-prepared affiliate. (Verified RB structure, masked: the borrower is a consolidated group of single-member LLC subsidiaries that own operating equipment and lease it to the parent; the audited notes consolidate an affiliate VIE — owned by the members and brought in via a debt guaranty — that owns equipment leased back to the borrower.)
- FR-CON-2 (P0). Surface related-party linkages discovered across statements and notes — large amounts due from related parties (RB ≈ \$11.4MM in 2023), related-party leases, and guarantees — and apply the configured treatment (receivables → non-current; debt → current unless subordinated; deduction from tangible net worth where covenants require). Credit signals, not items to bury in totals.
- FR-CON-3 (P0). Track an entity's legal name and any trade names / DBAs over time (the RB borrower adopted a new registered trade name in a recent year); a mid-history name change must not break period alignment or entity identity.
- FR-CON-4 (P1). Spread at entity level and roll up to a consolidated/combining view with eliminations where disclosed; track non-controlling interest and VIEs.
- FR-CON-5 (P2). Combine entities on different bases (GAAP vs. tax) with reconciliation notes.

7.13 Global cash flow and guarantor analysis (FR-GCF)

- FR-GCF-1 (P0). Spread guarantor PFS into a standardized personal cash-flow and net-worth view with liquidity haircuts and contingent liabilities.
- FR-GCF-2 (P0). Spread personal income (personal tax/W-2/K-1), distinguishing recurring from one-time and avoiding double-counting business distributions already captured at the entity.
- FR-GCF-3 (P0). Compute global cash flow / global DSCR: entity CFADS + net guarantor personal cash flow over combined debt service plus a living-expense factor — subtracting owner distributions — per a configurable definition.

- FR-GCF-4 (P1). Consolidate a guarantor's real-estate-owned schedule and outside interests; surface reliance on related-party debt repayment.

7.14 Validation, tie-outs, and exceptions (FR-VAL) — aggressive flagging

All checks are computed; failures are flagged with the offending cells and their sources and assigned a severity. Default thresholds are deliberately aggressive (surface liberally); alert fatigue is managed by severity, the ranked review queue, and bulk-acknowledge-with-reason — not by suppressing checks.

- FR-VAL-1 (P0). Balance control: Total assets = total liabilities + equity, per period, within tolerance (the RB spread carries a balance-control = 0 row).
- FR-VAL-2 (P0). Cross-foot: every subtotal equals the sum of its components; disagreement flagged as a likely extraction error (FR-EXT-7).
- FR-VAL-3 (P0). Cash-flow tie: ending cash on the cash-flow statement equals balance-sheet cash within a configurable rounding tolerance (the RB statements show \$1–\$2 ending-cash rounding in some years — a tolerance, not a dollar-exact fail).
- FR-VAL-4 (P0). Equity roll-forward: beginning equity + net income + contributions – distributions = ending equity.
- FR-VAL-5 (P0). Period continuity: prior-year ending balances equal current-year beginning balances.
- FR-VAL-6 (P0). Cross-document consistency: a figure appearing in multiple documents must agree (LOC balance in the debt note vs. balance sheet; comparative-year figures across two audits; application vs. audited statement). Disagreements are surfaced, not silently resolved; source precedence (FR-SRC-2, from the instructions file) chooses the primary.
- FR-VAL-7 (P0). Adjustment-consistency (FR-ADJ-4): an add-back applied in one period is flagged unless treated consistently across all periods — catches the RB bad-debt case.
- FR-VAL-8 (P0). Disclosure check: flag a disclosed obligation that is not recorded on the balance sheet (the RB ~\$573K) so it is not omitted from the credit view. (A pass-through tax-provision sanity check ships with tax handling in V1.)
- FR-VAL-9 (P1). Reasonableness / anomaly: period-over-period swings beyond bounds, sign flips, ratios outside plausible ranges, an unsupported improvement in leverage or coverage (leverage falls with no corresponding earnings/cash-flow change — often an add-back or reclass artifact), and RMA-peer divergence.
- FR-VAL-10 (P0). Severity levels on every issue — Critical (blocks finalize), High (analyst review), Medium (review before approval), Low (informational), Policy (requires approver override), Data (needs source/manual input) — and an aggregate status per spread (pass / pass-with-flags / blocked) that gates delivery.

7.15 SOP / policy configuration engine (FR-SOP)

- FR-SOP-1 (P0). Configure SCOA, reclassification rules, add-back library and caps, EBITDA/coverage definitions, custom metrics/templates, thresholds, period/LTM method, tolerances, and workflow as institution-level policy. Profiles are scoped (institution → product → deal) with inheritance/override; the per-package special-instructions file overrides the profile for that deal (FR-ING-3).
- FR-SOP-2 (P1) — V1. Required-documentation matrices by loan type and exposure (tiered assurance) and completeness gating.
- FR-SOP-3 (P0). Configure threshold rules (minimum DSCR, max LTV/leverage, debt-yield floors, normalized-NOI minimums) and evaluate them into pass/fail/exception flags — not decisions.
- FR-SOP-4 (P0). Exception & override policy: any deviation requires recorded rationale and routes for approval — RMA's rule that exceptions be “well supported and explained so the Credit Approver is aware of the change.”
- FR-SOP-5 (P1). Maker-checker configurable by exposure/risk; policy versioning with each spread recording the policy version that produced it.
- FR-SOP-6 (P2). Assist in translating a written spreading SOP into configured rules (Appendix C).

7.16 Human-in-the-loop review (FR-HIL)

- FR-HIL-1 (P0). A review surface ranking items by risk so reviewers spend time where it matters, not re-keying clean data.
- FR-HIL-2 (P0). Side-by-side source view: clicking any spread value highlights the exact source region (FR-EXT-5 coordinates); reverse lookup from source to spread.
- FR-HIL-3 (P0). Inline correction/override of any value, mapping, classification, or input; capture author, timestamp, superseded value, category, and rationale; re-propagate downstream.
- FR-HIL-4 (P0). Maker-checker workflow: analyst → reviewer → approver, with exceptions surfaced to the approver and a per-spread exception summary for the credit file.
- FR-HIL-5 (P1). Capture corrections as training signal (FR-MAP-6) without letting deal-level overrides silently change institution policy.

7.17 Audit trail and traceability (FR-AUD)

- FR-AUD-1 (P0). End-to-end lineage for every output value: source document + page + region → normalized SCOA value → reclassification (rule + footnote) → adjustment (provenance) → each metric it feeds. Queryable: “show how this was derived.”
- FR-AUD-2 (P0). Provenance status + confidence tier on every value — provenance enum: Reported / Derived / Policy-adjusted / Adjustment-candidate / Approved-adjustment / Assumption /

Override / Unsupported; confidence tier High / Med / Low. Both are visible in the UI and the export, and “Unsupported” can never silently populate a delivered metric.

- FR-AUD-3 (P0). Source citation on every delivered value (document + page/region) and generation of the source appendix (§7.18) — required depth, not key-values-only.
- FR-AUD-4 (P0). Record which Policy Profile, instructions file, ruleset, and metric/template versions produced each spread (reproducibility for model risk and exam); immutable log of all human actions.

7.18 Output, export, and integration (FR-OUT)

- FR-OUT-1 (P0). Clean Excel export — a single workbook with a clean presentation face and dedicated worksheets: Input tabs (normalized as-reported data) → Spread tabs (IS / BS / CF, common-size, multi-period + LTM) → Ratios & Custom Metrics (incl. covenant tests with pass/fail/headroom) → Exceptions & Inconsistencies → Commentary / Notes → Assumptions → Sources / Provenance. Live, inspectable formulas mirror the customer template (input tabs feeding the output tab) so the customer can audit and re-drive it.
- FR-OUT-2 (P0). Confidence & provenance in the export (High/Med/Low): cell notes on each value (confidence tier, source doc + page, provenance status) as the primary mechanism that keeps the face clean; a subtle traffic-light tier indicator (green/amber/red); provenance/assumption styling so assumptions and overrides are never mistaken for reported figures; an Exceptions worksheet listing every Low-confidence or flagged cell with a hyperlink, source, severity, and suggested action; and source hyperlinks from values into the Sources/Provenance tab and source appendix.
- FR-OUT-3 (P0). Support the Moody's-style reported / adjustment / adjusted presentation (separate adjustment column labeled by provenance) — matching the YETI strict spreads.
- FR-OUT-4 (P1). Borrowing-base output and, in the CRE phase, CRE/global-cash-flow outputs and MAA/YETI-style appendix binders with source pages behind spread tables.
- FR-OUT-5 (P1). API / integration: import the package and export spreads/metrics to the customer's LOS / credit platform / document store; expose borrower, entity, document, extracted value, period, line item, mapping, adjustment, metric definition/result, source locator, exception, review status, export package.
- FR-OUT-6 (P2). Live, refreshable borrower/portfolio views for monitoring.

8. Non-functional requirements

- NFR-1 Accuracy (parsing). Field-level extraction held to a benchmark standard (ParseBench-style: tables, charts, content faithfulness, semantic formatting, visual grounding). Sub-threshold fields route to review. Regression-tested against a labeled internal corpus.

- NFR-2 Explainability. No value is a black box; every number is reproducible from sources + formula and carries a provenance status and confidence tier.
- NFR-3 Auditability & retention. Immutable logs; document/extraction/mapping/formula/spread/export/approval versioning; reproducibility of any historical spread; examiner-ready exports.
- NFR-4 Security & privacy. Encryption in transit and at rest; RBAC; SSO/SAML/OAuth; document- and borrower-level permissions; tenant isolation; configurable data residency/retention; PII controls — borrower identity masked in shared artifacts/templates by default; no training on customer data without explicit permission; SOC 2-ready.
- NFR-5 Configurability. SCOA, rules, add-backs, metrics, thresholds, templates, and workflow are configuration; the instructions file overrides per package.
- NFR-6 Determinism. Same inputs + policy/instructions version ⇒ identical outputs; no hidden model-only calculations.
- NFR-7 Performance & scale. A standard multi-year + interim package spreads in minutes; concurrent multi-borrower throughput with queue/triage.
- NFR-8 Human-in-the-loop by design. The system proposes; the analyst/approver disposes. Confidence and exceptions are first-class.
- NFR-9 Observability & extensibility. Per-stage observability (confidence distributions, tie-out outcomes, override rates, drift vs. the golden set); new document types, SCOA lines, metrics, and templates add without core rework.

9. Accuracy, quality, and measurement

THE HEADLINE GUARDRAIL
Accuracy is never traded for automation rate; the headline guardrail is no silent low-confidence values.

9.1 ParseBench-aligned parsing KPIs

Dimension	What we measure	Target (initial)
Tables	Cell-structure fidelity on multi-year / merged-header tables	≥ 99% structure-correct
Numeric extraction	Per-cell value accuracy on financial tables	≥ 99.x%; 100% of sub-threshold cells flagged

Dimension	What we measure	Target (initial)
Content faithfulness	No hallucinated/transposed values; signage correct	≥ 99.9%
Semantic formatting	Hierarchy / subtotal / nesting preserved	≥ 99%
Visual grounding	Values resolvable to the correct source region	100% (hard requirement)

9.2 Spreading-specific KPIs

Classification accuracy (type/period/assurance) ≥ 98%; mapping accuracy with all reclassifications audit-noted; metric fidelity matching the customer's ground-truth template within rounding; tie-out pass rate 100% pass-or-flagged; LTM correctness 100% on reference cases; source-grounded coverage 100% for exported values; unsupported-metric silent-failure rate 0%. Straight-through-processing and time-to-spread are tracked as efficiency metrics, secondary to accuracy.

9.3 Golden set and known traps

Maintain a golden set from the reference packages (the RB working spread and the YETI strict spreads are machine-readable ground truth; the MAA and RB-credit-app images test OCR/vision). Gate releases on the §9.1 hard requirements. Permanent test cases: the LTM sign-label trap (Appendix D), the bad-debt add-back inconsistency, the footnote-only items (relief refund and unrecorded ~\$573K obligation), the cross-document comparative-figure discrepancy, the compilation that omits the cash-flow statement, and the interim-vs-year-end balance-sheet confusion.

10. Data model (key entities)

Package (folder + template + instructions; Borrower + Policy-Profile version) · Borrower (identity masked in shared artifacts) · Deal · Entity (borrower/subsidiary/VIE/affiliate/guarantor/property; legal name + trade names/DBAs over time) · Relationship (ownership/guaranty/related-party/consolidation) · Document (type, assurance, basis, currency, FYE, quality flags) · SourceValue ({document, page, bbox, raw_text, confidence, confidence_tier, provenance_status}) · Period (canonical key, type, fiscal cutoff, length-months, flow/stock, LTM method, component assurance) · CanonicalLineItem (template/SCOA-mapped; as-reported and as-spread layers) · Reclassification · Adjustment ({type, amount, provenance, recurring_flag, documentation, period_consistency_status}) · Assumption (first-class) · DisclosedObligation ({description, amount, recorded?, source_note}) · MetricDefinition (schema + DSL + bindings + threshold + version) → MetricValue (value, derivation, provenance, confidence_tier) ·

Instructions (source-precedence ranking + per-deal directives) · ValidationIssue (check, status, offending cells, severity) · ReviewAction · PolicyProfile · ExportPackage.

CONFIDENCE TIER AND PROVENANCE ENUM

Confidence tier: High / Med / Low. Provenance enum: Reported / Derived / Policy-adjusted / Adjustment-candidate / Approved-adjustment / Assumption / Override / Unsupported.

11. Nuances and edge cases (the details that decide credit)

- Partial-year / stub roll-forward done wrong: mixing flow and stock treatment, mismatched prior-year stub, or double-counting the overlap. Enforce $LTM_flow = CY_stub + PY_full - PY_stub$; $LTM_stock = \text{latest period-end}$.
- Mixed assurance within one LTM column (RB: audited prior-year anchor + compiled prior stub + internal current stub) — label per component.
- Cash-flow gap: a compilation/interim omitting the statement of cash flows means cash-flow/fixed-charge LTM metrics can't be tied without derivation — flag, don't fabricate.
- Footnote-driven items: contingent liabilities, litigation, subsequent events, and unrecorded obligations (the RB ~\$573K) live only in the notes.
- One-time income: relief refunds (the RB \$2.33MM), PPP forgiveness, grants, gains, insurance recoveries — tag non-recurring, keep out of run-rate.
- Pass-through / entity tax (V1): a borrower may carry no federal/state income-tax provision; model cash tax distributions rather than a book tax expense.
- Related-party items (RB ≈ \$11.4MM due from related parties; related-party leases): receivables → non-current, debt → current unless subordinated, TNW deduction where required.
- Consolidation / VIE: know what is consolidated vs. a separately-prepared affiliate; the RB borrower consolidates an affiliate VIE brought in via a debt guaranty, and subsidiaries lease operating equipment to the parent.
- Entity name changes / DBAs: a new trade name mid-history must not break entity identity or period alignment.
- Items hiding inside other lines: interest in COGS, D&A across COGS/opex, officers' comp in SG&A.
- Assumption-driven custom inputs: maintenance vs. growth capex, financed capex, cash vs. accrued interest, principal — labeled assumptions, never silently invented.
- Units, signs, scale; accounting basis (tax vs. GAAP vs. cash); restatements (keep both); interim balance ≠ year-end balance; cross-document discrepancies (comparative figures across two audits differ — surface and let precedence choose).

- CRE (V1): normalization minimums; in-place vs. market rent; stabilized vs. as-is NOI; rent-roll-to-OS reconciliation.
- Global cash-flow double counting: K-1/distributions already in business cash flow; rent in both property and guarantor — net out.
- Image-only / poor scans: skew, stamps, handwriting, multi-column — confidence routing and review are the safety net.

12. Risks and mitigations

Risk	Impact	Mitigation
Parsing errors on image/complex tables	Wrong metrics, silent failures	Benchmark-gated accuracy, confidence tiers, exception queue, visual grounding, internal tie-outs
Stub/LTM logic misapplied	Misstated cash flow & coverage	Flow/stock typing, enforced formula, derivation display, guardrails (FR-PER)
Cross-period adjustment inconsistency	Period-skewed EBITDA	Aggressive candidate-vs-approved + consistency check (FR-ADJ-4, FR-VAL-7)
Footnote-only items missed	Understated obligations	Footnote capture (FR-EXT-4), disclosed-obligation check (FR-VAL-8)
Over-automation erodes trust	Underwriters reject the tool	Human-in-the-loop, provenance + confidence, easy overrides
Template/instructions misread	Wrong structure or precedence	Template & instructions ingestion with hardcode detection; show resolved precedence
Alert fatigue from aggressive flagging	Reviewers ignore flags	Severity levels, ranked queue, bulk-acknowledge-with-reason
Sensitive data exposure / PII leak	Regulatory/legal	Encryption, RBAC, isolation; borrower identity masked in artifacts (NFR-4)
“Looks done but subtly wrong”	Bad credit decisions	Validation + severity gating, peer benchmarking, maker-checker

13. Phased roadmap

- MVP (P0) — C&I only. Inputs = folder + template + instructions; template ingestion as the anchor; extraction with confidence tiers + source coordinates + footnote capture + hardcode detection; assurance detection + instruction-driven source precedence; mapping to the template / RMA + tax-return SCOA with reclassifications and as-reported/as-spread layers; IS/BS/CF spread; multi-year + interim + LTM (assurance- and cash-flow-gap-aware); EBITDA/EBITDAR + add-back library (aggressive candidate-vs-approved + consistency) + one-time-income handling; standard ratios; custom-metric DSL + template ingestion + covenant-as-configured-metric + assumption inputs; multi-entity/related-party + VIE; global-cash-flow basics; validation + severity + provenance/confidence; SOP rules + instructions override; traceability + maker-checker + audit; clean Excel export with commentary worksheets, High/Med/Low confidence, and source appendix.
- V1 (P1) — CRE phase begins (multifamily first) + depth. CRE default templates (CREFC OSAR/NOIWS + agency multifamily), multifamily NOI/DSCR/LTV/debt-yield; pass-through/entity tax handling; completeness/freshness; borrowing-base/collateral; debt-schedule scenarios; peer benchmarking; consolidation roll-up; policy versioning + maker-checker; appendix binders; LOS/credit-platform API.
- V2 (P2) — breadth & monitoring. Other CRE types sequenced by market size (office, retail, industrial, hospitality); REIT FFO/AFFO depth; projections/pro forma; NL metric assist; covenant-definition parsing from loan agreements; live refreshable portfolio views; covenant monitoring.

MVP EXIT CRITERIA

Reproduce the RB C&I working spread (LTM Sep-2023 + custom EBITDA/EBITDAR/FCC tab) within rounding, fully traceable, with the bad-debt inconsistency and the unrecorded-tax/relief items flagged. (V1 exit: reproduce a multifamily property NOI/DSCR/LTV spread and the YETI borrowing-base / reported-adjustment-adjusted output.)

14. Resolved decisions (v0.4) and residual open items

Resolved (from the open-questions review)

- Inputs: one documents folder + spreading template + special-instructions file (the instructions carry source-priority and other guidance).
- MVP defaults: RMA C&I chart of accounts + tax-return structure; CRE defaults (CREFC OSAR/NOIWS + agency multifamily) ship with the CRE phase.
- Export confidence: clean Excel; confidence as High/Med/Low via cell notes + subtle traffic-light tier + provenance styling + Exceptions worksheet + source links.
- Flagging: aggressive by default (managed via severity + ranked queue + bulk-acknowledge).

- Source ranking: end-user-supplied via the instructions file (a configurable default is only a fallback).
- Export format: clean Excel workbook with dedicated commentary worksheets (Exceptions/Inconsistencies, Commentary, Assumptions, Sources/Provenance).
- Tax handling: deferred to V1.
- Domain priority: C&I first (MVP); CRE post-MVP, multifamily first (largest segment), other types by market size.
- Covenants: treated as configured custom metrics (definition + threshold + test period); loan-agreement parsing deferred.
- Source appendix: required on every delivered value (full depth, not key-values-only).
- PII: borrower identity masked in PRDs/templates/shared artifacts.

Residual open items

- ▶ Whether to add a Moody's-style chart of accounts to the default set (current MVP ships RMA + tax-return only).
- ▶ Exact confidence-tier thresholds (what counts as High vs. Med vs. Low) per document quality.
- ▶ Post-multifamily CRE ordering if office's distress should override pure market-size sequencing.
- ▶ First LOS/credit-platform integration target.

Appendix A — Standard chart of accounts: classification & GAAP-deviation rules

Defaults aligned to the RMA guide; configurable per institution and overridable by the customer template.

Assets

Source item	Spread as	Rationale
Prepaid expenses (insurance/rent/taxes/suppliers)	Non-current	Will not convert to cash to service debt
Restricted cash; sinking-fund cash	Non-current	Not liquid / not available
Cash surrender value of life insurance	Non-current	Not relied upon in normal course
Deposits (utility/insurance/lease/factor); long-term deferred charges	Non-current	Not a repayment source

Source item	Spread as	Rationale
A/R & notes receivable — related parties	Non-current	Discretionary repayment; if truly trade, keep as trade but flag
Notes receivable — third parties	Split current ≤ 12 mo / non-current	By maturity
Goodwill, patents, trademarks, capitalized costs	Intangible	Excluded from tangible net worth
Inventory (RM/WIP/FG), trade AR, marketable securities	Current	Normal liquidity

Liabilities & net worth

Source item	Spread as	Rationale
Current portion of long-term debt (CPLTD)	Current	Due within 12 months
Notes/mortgages/term debt — unrelated party	Current portion + non-current by maturity	Standard
Notes/mortgages/debt — related party	Current unless subordinated to the bank	May accelerate ahead of the bank
Accrued expenses (wages, interest, taxes, rent)	Current	Standard
Reserves (contingencies, self-insurance, taxes)	Current or non-current	Per expected payment timing
Subordinated debt	Non-current (may be equity-like per policy)	Subordination
Capital stock, surplus, retained earnings / member equity	Net worth	Equity

Income-statement break-outs: officers'/owner compensation; all D&A wherever it sits; interest embedded in COGS; split “interest expense and amortization of debt costs” into components.

Appendix B — Custom-metric examples (from the real RB customer spread; borrower masked)

House EBITDA family: CIT EBITDA = Gross Profit – SG&A – Rent; Adjusted EBITDA = CIT EBITDA + Adjustments (e.g., bad-debt expense, if approved/non-recurring); Adjusted EBITDAR = Adjusted EBITDA + Rent.

Leverage suite: Total Debt / CIT EBITDA; Total Debt / Adjusted EBITDA; (Debt + Leases) / EBITDAR.

Adjusted Fixed Charge Coverage — multi-step, with assumption-driven inputs

Numerator (A)

Component	Sign	Source / note
Adjusted EBITDA		House EBITDA + approved adjustments
Maintenance capex	–	Assumption: split of total capex; editable
Growth capex	–	Assumption: split of total capex; editable
Cash taxes (refund)	–	From cash-flow statement; ~0 for a pass-through
= Subtotal free cash flow		
Financed capex	+	Assumption: not clearly disclosed; editable
Lease / rent expense	+	From income statement: Rent
= Numerator (A)		

Denominator (B)

Component	Sign	Source / note
Cash interest		From cash-flow statement: actual interest paid
Principal repayments	+	From notes to FS: current portion of LTD
Refinanced debt / asset-sale amortization	–	Adjustments
Lease / rent expense	+	From income statement: Rent
= Denominator (B)		

Adjusted FCC = A / B (plus a “CIT FCC” variant on the CIT-EBITDA basis). Cash interest and principal come from the cash-flow statement and notes (verified cash interest paid was \$1.79MM and \$2.93MM in the two most recent years). Why financed capex is added back in the numerator: debt-financed capex did not consume operating cash, and the matching principal already sits in the denominator — so adding it back avoids double-penalizing coverage. The capex splits are labeled, editable assumptions, never reported figures.

Other illustrative custom metrics: Debt / Tangible Net Worth; Working-Capital Dependency = $(A/R + \text{Inventory} - A/P) / \text{Revenue}$; Distribution Coverage = $\text{Distributions} / \text{Net Income}$. CRE (V1): Debt Yield = NOI / Loan ; Max Supportable Loan = $\text{NOI} \div (\text{Min DSCR} \times \text{debt constant})$ at a test cap rate.

Appendix C — Representative lender SOP / policy excerpts

Illustrative of real customer environments; the SOP engine (FR-SOP) and the per-package instructions express rules like these.

- C.1 C&I spreading standard. “For exposure $\geq$ \$1.0MM, spread from CPA-prepared statements (audited preferred; reviewed acceptable); \$250K–\$1.0MM reviewed/compiled or tax returns; below that, tax/company-prepared. Minimum three fiscal years plus the most recent interim. Spread to the standard chart of accounts, which deviates from GAAP. Read the footnotes; related-party balances, contingent liabilities, and debt terms must be reflected. Independently verify before submission; document exceptions for the Credit Approver.” → FR-MAP, FR-SRC, FR-CON-2, FR-EXT-4, FR-HIL-4, FR-AUD; tiering → FR-SOP-2 (V1).
- C.2 EBITDA & add-back policy. “Permitted add-backs: D&A; interest; documented non-recurring items; restricted owner compensation. Each requires a recurring/non-recurring tag, applied consistently across all periods; period-specific treatment requires written justification. Disclosed unusual items (impairment, litigation, recall) are candidates for review, not approved add-backs. Government-relief items are non-recurring and excluded from run-rate earnings.” → FR-ADJ-2/3/4/7, FR-VAL-7, FR-OUT-3.
- C.3 Source precedence & assurance (end-user-driven). “Source precedence is supplied per package in the instructions file; absent that, use audited > reviewed > compiled > tax > company-prepared > internal interim. Flag stale spreads.” → FR-ING-3, FR-SRC-1..4.
- C.4 Cash-flow-gap policy. “If an interim omits a statement of cash flows, income-statement LTM may be computed, but cash-flow and fixed-charge metrics must be flagged unless supported by other schedules.” → FR-PER-8.
- C.5 Pass-through tax & disclosure (V1). “For pass-through/S-corp borrowers, do not record a federal income-tax provision; model owner tax distributions as a cash claim. Obligations disclosed in the notes but not recorded must be reflected.” → FR-ADJ-8, FR-EXT-4, FR-VAL-8.

- C.6 CRE NOI & sizing (V1). “Underwritten NOI normalizes vacancy (greater of actual or market minimum), management fee, replacement reserves; excludes one-time items; normalizes RE taxes to the reassessed basis for acquisitions. Min DSCR 1.20x–1.25x, debt yield $\geq$ 8–10%, LTV $\leq$ 75–80%.” → FR-CRE-1..5.
- C.7 Global cash flow & guarantors. “Owners $\geq$ 20% submit a PFS, personal returns, REO schedule, contingent-liability schedule. Global DSCR combines entity CFADS with net guarantor cash flow over combined debt service plus a living-expense factor; distributions are not double-counted.” → FR-GCF-1..4.
- C.8 Custom-metric governance. “Customer-defined metrics must be versioned, tested, approved, and traceable; a metric may be draft / analysis-only / approved / covenant / retired.” → FR-CUS-1/9/10. The Policy Profile bundles all of the above, scoped institution → product → deal, overridden per package by the instructions file.
- C.9 Why covenants live in the spread (the connection). A covenant is a custom metric + a threshold + a test period, and the spread is the engine that computes compliance. Scenario 1 (C&I): “Maintain Funded Debt / EBITDA $\leq$ 3.50x, where EBITDA = net income + interest + taxes + D&A + lender-approved non-recurring items, and Funded Debt excludes subordinated shareholder loans.” The covenant EBITDA and Funded Debt differ from the house versions, so the spread must compute the covenant-specific figures to test compliance at close and each quarter. Scenario 2 (CRE): “DSCR $\geq$ 1.20x on NOI underwritten at minimum 5% vacancy and \$250/unit reserves over actual debt service.” If the spread uses actual vacancy instead of the 5% floor, the covenant DSCR is wrong and a breach is missed. → FR-CUS-4.

Appendix D — Worked example: RB LTM Sept-2023 from partial-year audits (borrower masked)

The borrower provided full-year audited statements through FY2022 and nine-month interim statements for periods ended Sept-2022 (PY 9/22) and Sept-2023 (CY 9/23). The trailing-twelve-months view ending Sept-2023 is built per line for flow items as:

LTM (FLOW) FORMULA

LTM = CY 9/23 + (FY2022 – PY 9/22) — the current nine months plus the prior year's remaining quarter (Oct–Dec 2022).

Verified against the customer spread (USD thousands):

Line	FY2022	PY 9/22	CY 9/23	LTM Sep-23	Check: CY + (FY – PY)
Revenue	61,443	44,326	57,581	74,698	57,581 + (61,443 – 44,326) = 74,698
COGS	37,280	26,602	35,277	45,955	35,277 + (37,280 – 26,602) = 45,955
Adjusted EBITDA	10,082	6,580	11,971	15,473	11,971 + (10,082 – 6,580) = 15,473

Balance-sheet (stock) items take the most recent period-end — LTM cash, PP&E, and debt balances equal the Sept-2023 balances, not a roll-forward, and not the FY2023 year-end audit.

THREE NUANCES FOR ENGINEERING

(1) The customer's cell annotation reads “= CY 9/23 – (FY2022 – PY 9/22),” but the computed values correspond to CY + (FY – PY) (addition); the engine implements the verified arithmetic and shows the derivation rather than copying human shorthand. (2) The components carry different assurance — an audited prior-year anchor, a compiled PY 9/22 (which omits the cash-flow statement), and an internal CY 9/23 — so cash-flow-derived LTM metrics here must be flagged (FR-PER-8). (3) Footnote-only items (a \$2.33MM relief refund and a ~\$573K unrecorded tax obligation) and the one-year bad-debt add-back (\$3,662,933) sit behind these numbers and must be surfaced, not buried.

Appendix E — Source-document taxonomy → what we extract → where it feeds

Document	Key data extracted	Feeds
Audited/reviewed/compiled; 10-K/10-Q	IS, BS, CF, notes, segment data, contingencies	Core spread, reclassifications, add-backs, disclosed obligations
Business tax returns (1120/1120-S/1065/K-1)	IS/BS (tax basis), M-1/M-2, owner comp, ownership	Tax-basis spread, owner-comp break-out
Interim / stub statements	YTD IS, period-end BS, period length, assurance	Interim & LTM construction

Document	Key data extracted	Feeds
Debt schedule	Lender, balance, rate, payment, maturity, balloon, CPLTD, collateral, subordination	Interest, principal, coverage, leverage, scenarios
AR / AP aging	Buckets, >90-day, concentration, ineligible	Collateral quality, DSO, borrowing base
Borrowing-base certificate	Eligible AR/inventory, advance rates, reserves, availability	Borrowing base, availability
Rent roll (CRE)	Tenant, SF, rent, escalations, recoveries, term, options	EGI, occupancy, WALT, rollover
Operating statement / T-12 (CRE)	GPR, vacancy, other income, opex, NOI	NOI/NCF build
Appraisal (CRE)	As-is/as-stabilized value, cap rate, comps	Value, LTV, cap-rate reconciliation
Guarantor PFS; personal tax/W-2/K-1	Assets (liquidity), liabilities, net worth, income	Global cash flow, guarantor net worth
Org / ownership docs; instructions file	Entity structure, ownership %, trade names; source precedence + directives	Consolidation, precedence, per-deal config

Appendix F — Glossary

- SCOA — standard chart of accounts.
- CPLTD — current portion of long-term debt.
- EBITDA(R) — earnings before interest, taxes, D&A (and rent).
- CFADS — cash flow available for debt service.
- DSCR / FCCR — debt-service / fixed-charge coverage ratio.
- NOI / NCF — net operating income / net cash flow (CRE).
- EGI — effective gross income.
- LTV / LTC — loan-to-value / loan-to-cost.
- Debt yield — NOI / loan.
- Cap rate — NOI / value.
- LTM / TTM — last / trailing twelve months.
- Stub / interim — a partial fiscal period (e.g., 9 months).
- WALT — weighted-average lease term.

- VIE — variable-interest entity.
- EBITDAre — NAREIT-defined EBITDA for real estate.
- FFO / AFFO — funds (adjusted funds) from operations.
- CREFC OSAR / NOIWS — CRE Finance Council Operating Statement Analysis Report / NOI Adjustment Worksheet (the CRE NOI-normalization standard).
- Confidence tier — High / Med / Low per delivered value.
- Provenance status — Reported / Derived / Policy-adjusted / Adjustment-candidate / Approved-adjustment / Assumption / Override / Unsupported.
- Policy Profile — the versioned configuration bundle encoding a lender's spreading SOP.
- Instructions file — per-package end-user directives, including source precedence.

Appendix G — Export workbook structure (clean Excel)

A single workbook; clean presentation face, with confidence and issues corralled into notes and dedicated tabs.

Worksheet	Contents
Income Statement	Standardized IS spread, multi-year + interim + LTM, common-size %, YoY growth (live formulas)
Balance Sheet	Standardized BS spread with reclassifications, common-size %
Cash Flow	Statement of cash flow / UCA cash flow; reconciliation of net worth
Ratios & Metrics	Standard ratios + custom metrics + covenant tests (value, threshold, pass/fail, headroom)
Inputs	Normalized as-reported data feeding the spread tabs
Exceptions & Inconsistencies	Every flagged item: description, severity, affected cell (hyperlink), source, suggested action
Commentary / Notes	Analyst/underwriter narrative, judgment notes, open questions
Assumptions	Every assumption (e.g., capex splits) with value, basis, owner — styled distinctly
Sources / Provenance	Per value: source doc + page/region (hyperlink), provenance status, confidence tier

On-cell treatment: each delivered value carries a cell note (confidence tier High/Med/Low + source + provenance); a subtle traffic-light indicator for the tier; assumptions and overrides styled distinctly from reported figures; values hyperlink to the Sources/Provenance tab and source appendix. The face stays clean and audit-ready; nothing low-confidence is written silently.

Prepared for team review (v0.4). Grounded in the RMA Guide to Spreading Financial Statements (2022), the CREFC Investor Reporting Package (OSAR/NOIWS), Fannie Mae / Freddie Mac multifamily standards, ParseBench (arXiv 2604.08538), the CFI commercial-credit sample cases, and the real customer (RB) and public-company (MAA, YETI) spread sets — RB facts verified against the borrower's audited statements, with borrower identity masked.